Subodh Wagh

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Professional Summary

M.Sc. Computer Science graduate with interests in backend development, systems programming, distributed systems, and network infrastructure. Developed concurrent network services, compiler toolchains, and programmable data-center network emulators using C, Go, Python, and Linux networking primitives.

Technical Skills

Languages: C, Go, Python, Bash, x86 Assembly
Databases: PostgreSQL
Systems: POSIX APIs, Multithreading, TCP/IP Sockets, Process Synchronization, IPC, Memory Management
Networking: TCP/IP, BGP, EVPN, VXLAN, VRFs, Linux Network Namespaces
Tools: Git, FRRouting (FRR), NeST Emulator, Docker, Wireshark, tcpdump, GDB

Projects

Concurrent In-Memory KV Cache Server *View Code*
Go, TCP Sockets, Concurrency

- Implemented a concurrent in-memory key-value cache server in Go using TCP sockets and a custom binary protocol.
- Designed a 256-shard concurrent storage engine to reduce lock contention under parallel workloads.
- Sustained **70,560 requests/second** while serving **1,000 concurrent clients**, using deterministic lock ordering to maintain consistency during multi-key operations.

EVPN-VXLAN Data Center Fabric Emulator *View Code*
Python, FRRouting, NeST Emulator, Linux Networking

- Developed Python automation to provision and configure EVPN-VXLAN Spine-Leaf data-center topologies using the NeST network emulator.
- Provisioned a virtual fabric comprising **20 spine routers**, **10 leaf routers**, and **10 hosts** using Linux network namespaces and virtual Ethernet (veth) links.
- Automated FRRouting configuration for **200 underlay BGP peering relationships**, enabling EVPN-VXLAN overlay connectivity across the fabric.

Recursive-Descent Compiler *View Code*
C, x86-64 Assembly

- Implemented a lexical analyzer and recursive-descent parser for a custom programming language.
- Implemented symbol table management and semantic analysis for identifier resolution and scope checking.
- Generated x86-64 assembly for variables, arithmetic expressions, conditional statements, and loops.

Two-Pass x86 Assembler *View Code*
Python, x86 ISA

- Built a two-pass assembler to resolve forward label references and generate machine code.
- Implemented instruction encoding, ModR/M byte generation, and relative address calculation.
- Validated generated machine code against NASM output for instruction encoding correctness.

Education

M.Sc. Computer Science Savitribai Phule Pune University
CGPA: 7.93 / 10
Coursework: Operating Systems, Computer Networks, Distributed Systems, Compiler Design

B.Sc. Computer Science PVP College, Ahilyanagar
CGPA: 8.60 / 10

Leadership

- **Core Member, CSFG Cell (SPPU):** Organized technical seminars and departmental events as a Core Member of the CSFG Cell.